Capstone Project
Stock Market Anomaly Detection

1. Overview

Goal. Detect *unusual* market days and stock days using only daily price and volume data. "Unusual" means very large moves or spikes or dip, unusually high volume, or unusually wide ranges *compared to recent history*. No labels, no news, no complex models.

Why this matters. Markets often have stress days. Spotting them helps:

- **Monitoring:** know when the whole market is stressed (*market anomaly*).
- **Diagnostics:** know when a single stock behaves differently from others (*stock anomaly*).
- **Learning:** practice time-series hygiene (avoid leakage), rolling statistics, and simple unsupervised rules.

What you will build.

1. A tiny feature set (return z-score, volume z-score, intraday range percentile), computed with *rolling windows* that use only the past.
2. A simple *rule-based detector*: if today's feature is extreme, flag an anomaly and label it as *crash*, *spike*, or *volume_shock*.
3. A **Date Query**: input a date and print the market status and the list of anomalous tickers for that day.
4. A **Monthly Mini-Report**: one table summarizing abnormal dates and why they were flagged.

Educational analytics only — not investment advice.

What exactly counts as "unusual"?

We call a day *unusual* when today's behaviour is extreme *relative to its recent history*. We measure "recent history" using simple **rolling windows** (past data only; no look-ahead):

- **Rolling windows (defaults):** returns $W_{\text{return}} = 63$ trading days, log-volume $W_{\text{volume}} = 21$ days, intraday range $W_{\text{range}} = 63$ days.
- **Warm-up:** do not score a day until you have at least the needed past days (`min_obs` $\geq \max\{W_{\text{return}}, W_{\text{volume}}, W_{\text{range}}\}$).

Per-stock indicators (computed for each ticker i on day t)

$$\text{Return: } r_{i,t} = \frac{\text{AdjClose}_{i,t} - \text{AdjClose}_{i,t-1}}{\text{AdjClose}_{i,t-1}}$$

$$\text{Return z-score: } \texttt{ret_z}_{i,t} = \frac{r_{i,t} - \mu_{i,t}^{(r)}}{\sigma_{i,t}^{(r)}} \quad \text{where } \mu_{i,t}^{(r)}, \sigma_{i,t}^{(r)} \text{ are mean/std over } [t - W_{\text{return}}, t-1]$$

$$\text{Volume z-score: } \texttt{volz}_{i,t} = \frac{\log V_{i,t} - \mu_{i,t}^{(\log V)}}{\sigma_{i,t}^{(\log V)}} \quad \text{using } [t - W_{\text{volume}}, t-1]$$

$$\text{Intraday range percentile: } \texttt{range_pct}_{i,t} = \text{percentile of } \frac{\text{High} - \text{Low}}{\text{Close}} \text{ vs the last } W_{\text{range}} \text{ days in } [t - W_{\text{range}}, t-1].$$

Unusual (stock-day) = any simple trigger fires

$$\left|\texttt{ret_z}_{i,t}\right| > 2.5 \quad \text{or} \quad \texttt{volz}_{i,t} > 2.5 \quad \text{or} \quad \texttt{range_pct}_{i,t} > 95\text{th percentile}.$$

Type label:

- **crash** if $r_{i,t} < 0$ and $\left|\texttt{ret_z}_{i,t}\right| > 2.5$
- **spike** if $r_{i,t} > 0$ and $\left|\texttt{ret_z}_{i,t}\right| > 2.5$
- **volume_shock** if $\texttt{volz}_{i,t} > 2.5$ (can co-exist with crash/spike)

Market context (same day, across your small universe)

$$r_t^{\text{mkt}} = \text{mean of } r_{i,t}, \qquad \text{breadth}_t = \frac{\#\{i : r_{i,t} > 0\}}{\text{tickers}}.$$

Unusual (market-day) if $|r_t^{\text{mkt}}|$ is above its own 95th percentile (computed over $[t - W_{\text{return}}, t-1]$) *or* $\text{breadth}_t < 0.3$.

Severity (0–100) — optional but useful A simple severity score combines the "how extreme" signals:

$$\text{severity}_{i,t} = \text{avg}\Big(\text{percentile}\big(|\texttt{ret_z}_{i,t}|\big),\ \text{percentile}\big(\texttt{volz}_{i,t}\big),\ \texttt{range_pct}_{i,t}\Big) \times 100.$$

2. Dataset (Kaggle link)

Daily NASDAQ OHLCV CSVs (one per ticker) plus a metadata file. Use **Adj Close** for returns.
Link: https://www.kaggle.com/datasets/jacksoncrow/stock-market-dataset

Columns: `Date`, `Open`, `High`, `Low`, `Close`, `Adj Close`, `Volume`. Data ends **01-Apr-2020**.
Tip: Returns based on `Adj Close` account for splits/dividends.

3. Minimal Scope (keep it small)

- **Tickers:** QQQ + any 5–10 liquid stocks (e.g., AAPL, MSFT, AMZN, NVDA, META).
- **Dates:** Use the dataset as provided.
- **Splits:** Train=2018, Validation=2019, Test=2020-Q1 (Jan–Mar).

4. Theory & Concepts

4.1 What is an anomaly?

An **anomaly** is an observation that looks unlikely relative to recent behavior.

- **Point anomaly:** one day that is extreme (e.g., big jump or crash).
- **Contextual anomaly:** extreme relative to context (e.g., a 2% move might be normal in volatile months but unusual in calm months).

4.2 Rolling windows and leakage

Rolling window means: when deciding day t, compute statistics using *only* days $t-1, \ldots, t-k$.
Leakage means using information from t or the future to decide t.

4.3 Z-scores (standardized surprise)

For a series x_t, rolling mean μ_t, std σ_t:

$$z_t = \frac{x_t - \mu_t}{\sigma_t}.$$

4.4 Intraday range percentile

(High − Low)/Close compared to the last 63 days.

4.5 Market status (breadth)

$$r_t^{\text{mkt}} = \text{mean of } r_t, \quad \text{breadth}_t = \frac{\#\{r_t > 0\}}{\text{tickers}}.$$

Methodology (Clustering-only: K-Means & DBSCAN)

A. Leakage-safe features

For each ticker i on day t compute features using only the past:

$$\begin{aligned}
\texttt{ret_z}_{i,t} &= \frac{r_{i,t} - \mu_{i,t}^{(r)}}{\sigma_{i,t}^{(r)}} \quad \text{with } (\mu_{i,t}^{(r)}, \sigma_{i,t}^{(r)}) \text{ from } [t-63, t-1], \\
\texttt{volz}_{i,t} &= \frac{\log V_{i,t} - \mu_{i,t}^{(\log V)}}{\sigma_{i,t}^{(\log V)}} \quad \text{with } (\mu_{i,t}^{(\log V)}, \sigma_{i,t}^{(\log V)}) \text{ from } [t-21, t-1], \\
\texttt{range_pct}_{i,t} &= \text{percentile of } \frac{\text{High} - \text{Low}}{\text{Close}} \text{ vs } [t-63, t-1].
\end{aligned}$$

Warm-up: do not score until each window has enough past points.

B. Scaling and design matrix

Build $X = \{(\texttt{ret_z}, \texttt{volz}, \texttt{range_pct})\}$ stacked across tickers and days. Fit a standard scaler on the **training set** only (2018) and apply it to Validation (2019) and Test (2020-Q1).

C. Split and walk-forward

Train=2018, Validation=2019, Test=2020-Q1. Hyperparameters are chosen on Train/Val and then *frozen* for Test. Optionally refit models on an *expanding* history up to $t-1$ (same hyperparameters), then score the next block.

D. K-Means anomaly detection

Fit (Train). Choose k using elbow or silhouette on training X. Fit K-Means on standardized features.

Score and threshold. For a standardized point x, let $c(x)$ be its nearest centroid and $d(x) = \|x - c(x)\|_2$. Compute the distribution of $d(\cdot)$ on **training points** within each cluster and take the q-th percentile (e.g., $q = 97.5$) as the cluster threshold τ_c.

$$\text{K-Means anomaly if } d(x) > \tau_{c(x)}.$$

Validation. Tune (k, q) on 2019 to keep total flag rate near 2–8% and to maximize overlap with top $|r|$ days.

E. DBSCAN anomaly detection

Fit (Train). Pick `min_samples` (e.g., 10–20) and set `eps` from the k-distance elbow (with k = `min_samples`) on the training set. Fit DBSCAN on standardized training points. Points labeled -1 are *noise*.

Score (Val/Test) with walk-forward. Before each scoring block (e.g., monthly), refit DBSCAN on an expanding window up to $t - 1$ using the fixed (`eps`, `min_samples`). Then run DBSCAN on that window *plus* the new block and take labels for the new block; label -1 as **anomaly**.

Validation. Adjust (`eps`, `min_samples`) on 2019 to hit the target flag rate and produce sensible clusters.

F. Hybrid and consensus (optional)

Produce two scores per point: K-Means distance $d(x)$ and DBSCAN label $\in \{-1, \text{cluster}\}$. You may report:

- **union**: anomaly if K-Means *or* DBSCAN flags (higher recall),
- **intersection**: anomaly if both flag (higher precision).

G. Market layer (daily)

For each date t, aggregate across tickers:

$$\text{flag_rate}_t = \frac{\#\{\text{tickers flagged on } t\}}{\text{tickers}}, \quad r_t^{\text{mkt}} = \text{mean of } r_{i,t}, \quad \text{breadth}_t = \frac{\#\{r_{i,t} > 0\}}{\text{tickers}}.$$

Use flag_rate as a simple *market stress indicator* and report $(r_t^{\text{mkt}}, \text{breadth}_t)$.

H. Evaluation without labels

- **Flag rate control:** hold overall anomaly fraction near 2–8%.
- **Tail overlap:** fraction of flagged points that fall in the top p% of $|r|$ (e.g., $p = 2$ or 5).
- **K-Means quality:** silhouette score on Train; stability of centroids month to month.
- **Case studies:** inspect largest K-Means distances and DBSCAN noise cases; verify explanations via feature values.

I. Serving

- **Date Query:** for date t, list all tickers flagged by K-Means and/or DBSCAN with their distances and neighborhood counts.
- **Monthly report:** abnormal dates, top spikes/dips by return, breadth, and market stress (flag_rate).

6. Required Outputs

A. Daily Anomaly Card (CSV)

```
date,ticker,anomaly_flag,type,ret,ret_z,volz,range_pct,why
2020-02-27,AAPL,1,"crash + volume_shock",-0.042,-2.9,2.8,0.035,"|ret_z|>2.5; volz
    >2.5"
```

B. Market-Day Table (CSV)

```
date,market_ret,breadth,market_anomaly_flag
2020-02-27,-0.031,0.12,1
```

C. Date Query

Input: `-date YYYY-MM-DD`; Output: market status + anomalous tickers.

D. Monthly Mini-Report

Date	Ticker	Type	ret_z	volz	Mkt flag	Why
2020-02-27	AAPL	crash+volume_shock	-2.9	2.8	1	$\lvert$`ret_z`$\rvert > 2.5$, `volz`> 2.5
2020-02-27	MSFT	crash	-2.7	2.1	1	$\lvert$`ret_z`$\rvert > 2.5$

7. Train/Val/Test

Train=2018, Val=2019 (tune thresholds), Test=2020-Q1 (lock thresholds, target 2–8% flag rate).

8. Deliverables & Grading

Repo + CSVs + README. : Overview & theory, Cleaning, Features, Detector, Query & outputs, Mini-report.

9. Quick Start

```
python -m src.walkforward --universe QQQ,AAPL,MSFT,NVDA,AMZN,META
python -m src.query --date 2020-02-27
python -m src.monthly --month 2020-02
```